



Night sight

At night, lions' eyes seem to glow when they are caught by a beam of light. This effect is common in nocturnal mammals and also in crocodiles and sharks. It is caused by a reflective layer inside the eye called the tapetum lucidum, which bounces light back through the retina, helping animals to see in the dark. Eyes also adjust to the dark by chemical changes that make them better at sensing dim light.



Beady gaze

Predators often have forward-facing eyes, which are good at judging depth, while plant-eaters tend to have eyes on the sides of their heads, which lets them see danger from all around. Chameleons get the best of both worlds. Their eyes can swivel independently to look all around, but they face forward when the chameleon focuses on its prey.

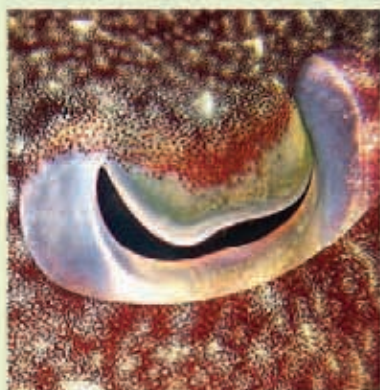
Camera vision

Clinging to a branch, a tarsier sees its rain-forest home through giant eyes. Like those of all other vertebrates, its eyes work like cameras, focusing light on a screen called the retina, which is packed with light-sensitive cells. During the day, its pupils close up to stop it from being dazzled by bright lights. At night, when it hunts, they open wide, letting in as much light as possible.

Pupil narrows in daylight



Amazing eyes



Invertebrate vision

Cuttlefish have a single lens and a retina, like the eyes of mammals and other vertebrates, but have W-shaped pupils.



Wipe clean

Red-eyed tree frogs have a mottled third eyelid, or nictitating membrane, which cleans the eye like a windshield wiper.



Lashes of color

Ground hornbills' eyelashes screen their eyes from the Sun's glare and make the eyelids shut instantly when touched.



Seeing through a slit

Tokay geckos have slit-shaped pupils in a speckled iris, which camouflages the eye. After dark, their pupils open wide.